ps1-av-hat, one channel. TVP7002 digitises the PS1 analog RGB (16-bit Y/C bus) into the CH32H417 dev board via two 3x19 headers, a PCM2903C presents the line audio as a USB Audio Class device on its own port.

Input

File: input.kicad_sch

TVP7002

File: tvp7002.kicad_sch

Power

File: power.kicad_sch

Audio

File: audio.kicad_sch

MCU

File: mcu.kicad_sch

rixnobis

Sheet: /

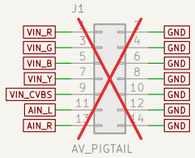
File: ps1-av-hat.kicad_sch

Title: ps1-av-hat: one-channel PS1 RGB + audio hat for CH32H417

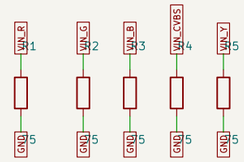
Size: A3 | Date: | **Rev: A**

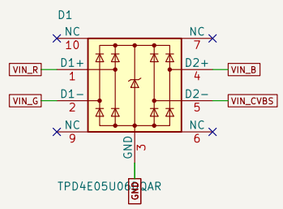
KiCad E.D.A. 10.0.7 | Id: 1/6

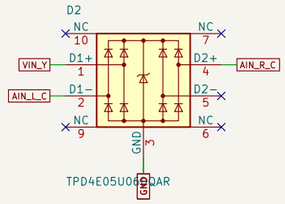
PS1 AV multi-out via a short pigtail to a multi-AV male plug (pinout in the FX3 project docs/design.md). Multi-out +5V (pin 3) and S-Video C (pin 6) are not carried.

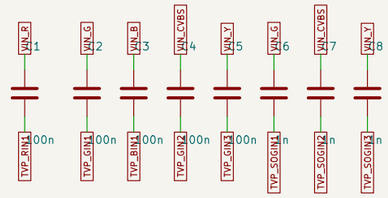


75 ohm termination at the connector, then ESD, then AC coupling (TVP7002: 0.1 uF; SOG inputs: 1 nF). CVBS and Y also reach spare ADC inputs, so luma can be digitised.

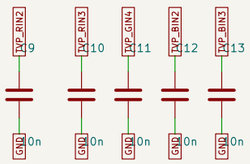








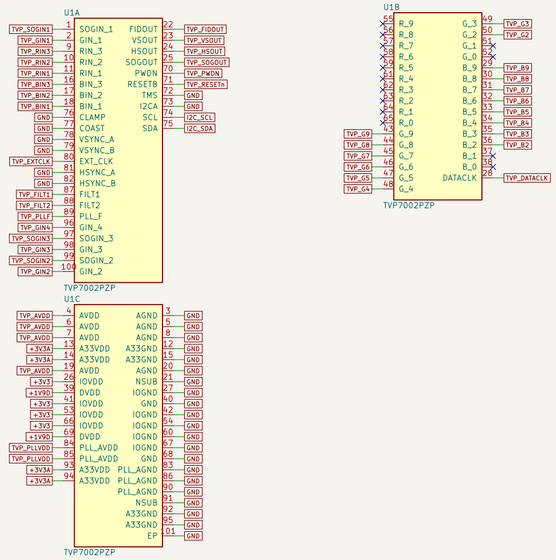
Unused TVP7002 analog inputs: 10 nF to GND (SLES206C Table 1).



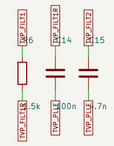
rixnobis		
Sheet: /Input/		
File: input.kicad_sch		
Title: AV multi-out input, termination, ESD		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/6

TVP7002 on 3.3 V IOVDD. The H417 VIO18 rail MUST be 3.3 V (Maker go R39 fitted 0R, or firmware PWR_CTLR VIO18=3.3V) before RESETB is released: TVP IOVDD is 3.3 V and the VIO18 pins are not 5V/over-voltage tolerant.

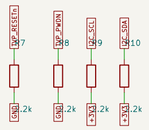
Sync comes from SOGIN (composite or luma; the PS1 has no sync pin), so the digital HSYNC/VSYNC inputs are grounded as unused. I2C address B8h (I2CA low).



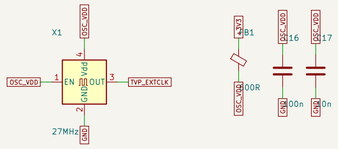
H-PLL loop filter (SLES206C Figure 4): 1.5k + 100n FILT1-PLL_F, 4.7n FILT2-PLL_F.



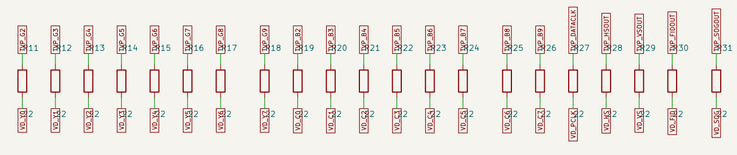
Control. RESETB is held low by 2.2k until the CH32H417 releases it (TVP_RESETn is an MCU GPIO, PA4), so the TVP7002 stays in reset until firmware is up (SLES206C: no I2C activity before reset completes). PWDN 2.2k to GND (Figure 8). I2C bus: 2.2k pull-ups to +3V3.



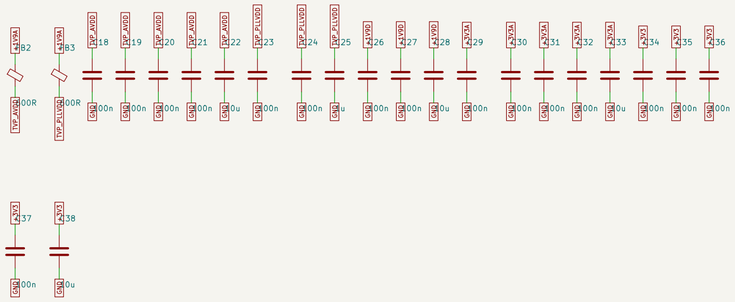
27 MHz reference on EXT_CLK for mode detection (SLES206C recommends 27 MHz; the internal reference is about 6.5 MHz and imprecise).



Output bus: 22R series at the source, only the 16 bits used plus clock and syncs. G[9:2] -> VD_Y0..7 (luma), B[9:2] -> VD_C0..7 (chroma); R[9:2] are unconnected.



Supply decoupling. SLES206C gives no values; this follows the OSSC: ferrites on AVDD and PLL_AVDD, 100n per pin, bulk per rail.

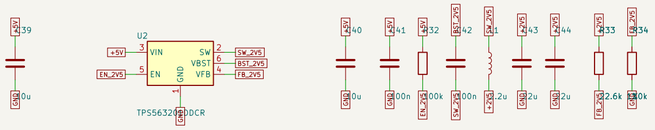


rixnobis
Sheet: /TVP7002/
File: tvp7002.kicad_sch

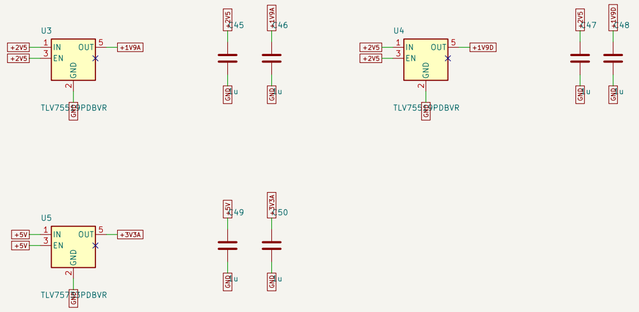
Title: TVP7002 RGB digitiser

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.7		Id: 3/6

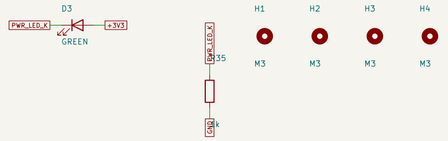
Power comes from the dev-board headers: +5V (J4 pins 7-9) and +3V3 (J4 pins 10-12); no USB 3 connector. +3V3 feeds TVP7002 IOVDD, the oscillator, pull-ups and the bus-control gate. +5V feeds one TPS563201 buck to 2.5 V (SLVSD90B Table 7-2: R2 = 10k, L = 2.2 uH, 2 x 22 uF out), which is the pre-regulator for the 1.9 V LDOs (OSSC pattern). EN has an internal pull-down, so it is tied up through 100k.



LDOs: 2.5 V -> 1.9 V for TVP7002 AVDD/PLL and DVDD (two TLV75519, 500 mA); +5V -> 3.3 V for TVP7002 A33VDD alone (TLV75733, 1 A), kept off the board's 3V3 rail.



Power LED on +3V3, mounting holes.

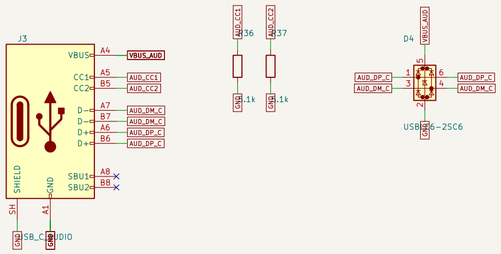


Power flags (ERC): +2V5, +3V3, +5V, GND, OSC_VDD, TVP_AVDD, TVP_PLLVDD, VBUS_AUD

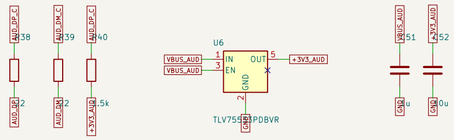


rixnobis	
Sheet: /Power/	
File: power.kicad_sch	
Title: Power from the dev-board headers	
Size: A2	Date:
KiCad E.D.A. 10.0.7	Id: 4/6

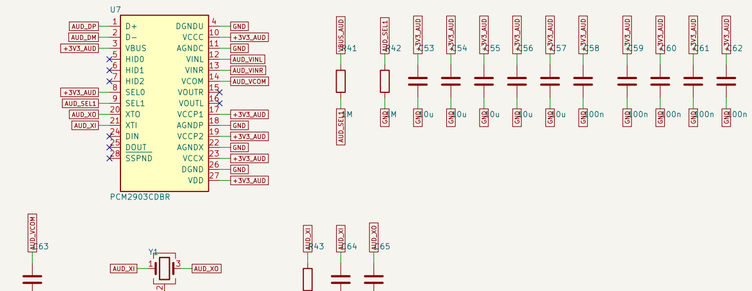
PCM2903C on its own USB 2.0 Type-C port. TI documents the PCM2903C self-powered only (SBFS038A Figure 41), so a 3.3 V LDO makes its supply from this port's VBUS.

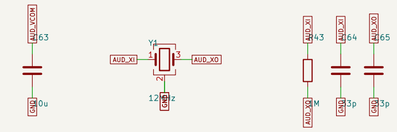


D+/D−: 22R series; 1.5k D+ pull-up on the connector side (Figure 41). TI gates the pull-up with an AND of VDD and VBUS; here VDD only exists when VBUS does, so it goes to +3V3_AUD.



PCM2903C wiring per Figure 41: VBUS pin and SEL0 to VDD; SEL1 to the port VBUS with 1M to GND; 10 uF on every supply pin and VCOM; 12 MHz crystal with 1M across it. DAC, S/PDIF and HID pins unused.





Line input: 2.2 uF coupling, then 10k into the 30k ADC input (x0.75, so 2.6 Vpp maps to the 1.98 Vpp full scale) with 330 pF to GND (anti-alias pole ~48 kHz).

rixnobis

Sheet: /Audio/
File: audio.kicad_sch

Title: PCM2903C USB audio

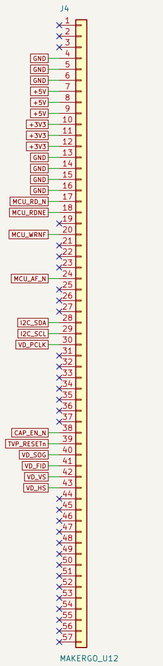
Size: A2	Date:	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 5/6

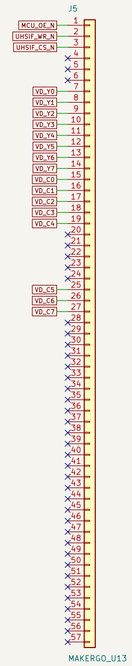
J4: Maker go CH32H417QEU6 header U12
(3x19, 2.54 mm, pins 1..57 column-wise).
Pin-to-port table in gen_hat_sch.py.

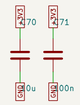
J5: Maker go CH32H417QEU6 header U13
(3x19, 2.54 mm, pins 1..57 column-wise).
Pin-to-port table in gen_hat_sch.py.

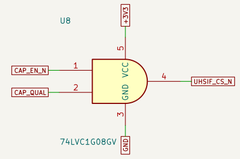
+3V3 bulk at the header: 10u + 100n.

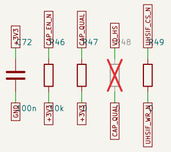
Bus control. UHSIF CS# and WR# (active low) are driven low together while capturing: Y = CAP_EN_N
AND CAP_QUAL. CAP_EN_N idles high through 10k; CAP_QUAL is tied high by a 0R, with a DNP 0R to VD_HS
as an option to gate on HSOUT later. The 0R CS#-WR# link lets them be split.



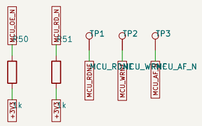








MCU_OE_N and MCU_RD_N: 1k pull-ups to +3V3 (the board loads OE# with an LED to VDDIO). MCU_RDNE,
MCU_WRNF, MCU_AF_N: test points only.



rixnobis

Sheet: /MCU/
File: mcu.kicad_sch

Title: CH32H417 dev-board headers and bus control

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.7		Id: 6/6